

Ritesh Kadmawala

Founder, Vertexcover Labs · Multi-agent systems, model fine-tuning, agentic delivery

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SUMMARY

15+ years building products and startups. Two companies cofounded, VP Engineering at InVideo, backend lead for risk and security at Xapo, and principal consultant at Rephrase.ai (acquired by Adobe), Notch and Dunzo. I now run Vertexcover Labs, a small fully remote team that builds AI products for funded startups. Most of that work is multi-agent systems, model fine-tuning and evals. We also build and open source the harness our own delivery runs through, which takes a feature from spec to PR with review, hard quality gates, and a QA agent that checks it actually works. I run two to four engagements at a time.

EXPERIENCE

Founder — Vertexcover Labs

Jan 2018 – Present · Remote

Small fully remote engineering team building AI products for funded startups.

- **Multi-agent systems in production:** orchestration and planning, tool calling, agentic RAG, evals and tracing, guardrails, human-in-the-loop review. Shipped for Notch and Theo (South Park Commons), Arintra (YC), Springworks/Albus, pCloudy by Opkey, rocket.new and DhiWise.
- **Model fine-tuning and evaluation** where a general model is not good enough. Fine-tuned the reasoning model behind Arintra's autonomous medical coding agent for **10% higher accuracy at 50% lower cost**, with benchmarking, vLLM optimization and a Langfuse eval pipeline behind it. Agentic RAG for Springworks/Albus lifted resolved queries 25%.
- Built **Harness Engineering**, our open-source Claude Code and Codex plugin. Spec to PR in isolated worktrees, EARS acceptance criteria, dependency-ordered phase plans, parallel coding agents under TDD, and pass/fail gates on typecheck, lint, tests and coverage.
- Own technical strategy and delivery across two to four concurrent engagements: scoping, architecture, the spec, and the engineers who ship against it. Reviewing agent diffs for hallucinated APIs, unsafe defaults and scope creep is part of the job.

Principal Consultant — Notch (usenotch.ai)

Sep 2024 – Present · Remote

Seed-stage (\$8M, South Park Commons / Wing) AI ad agent that researches, scripts, generates and edits complete ads from a brief.

- Lead AI and backend: multi-agent orchestration, RAG, tool calling, workflow design, tracing, evals, guardrails, CI/CD and production hardening.
- Built the video ad agent, from research and script through generation, assembly and cut, plus remix that produces new hooks, CTAs and cuts off an ad already performing.
- Shipped v1 in two weeks, ad-creation time down about 50% against manual production. The platform now generates 5,000+ ads and supports \$5M+ in managed ad spend.

Principal Consultant — Rephrase.ai (acquired by Adobe)

Oct 2021 – Nov 2023 · Remote

Generative text-to-video platform; Adobe's first generative-AI acquisition.

- Built the browser-based rendering engine so preview and final render run the same code path. Features got built once instead of twice, and the drift between preview and output went away.
- Re-architected the video pipeline with benchmarking, concurrency, GPU scheduling, queueing theory and custom rate limits: **4x faster generation, 75% lower hardware cost, 100K videos/day**, consistent SLA across customer tiers.
- Raised team TypeScript proficiency through code-review standards, testing protocols and weekly tech talks.

Fractional CTO — Ecel.ai

Dec 2023 – May 2024 · Remote

Led AI-powered ad creative generation: stack architecture, team mentoring, and hands-on delivery alongside the advisory work.

Engineering Consultant — Dunzo

Jun 2022 – Dec 2022 · Remote

- Re-architected the delivery pricing rule engine, offer and coupon search and filtering, bulk coupon generation to 5M codes, and repeat-abuse fraud prevention.
- Custom error-handling library that tripled bug detection and resolution efficiency. Introduced trunk-based development on the projects I led.

Lead Software Consultant — Pashi

Jan 2021 – Sep 2021 · Remote

No-code platform to build, test, simulate, monitor and operate manufacturing production lines, running at some of India's largest auto makers.

VP Engineering — InVideo

May 2020 – Jan 2021 · Remote

- Led 20+ engineers across multiple teams at a consumer video creation and editing platform.
- Rebuilt devops and infrastructure with Jenkins and Terraform, plus monitoring, logging and error tracking. Set up hiring, release, documentation and code-review processes from scratch.
- Ran programmes across personalization, analytics, audio editing and a UI/UX revamp.

Backend Lead, Risk & Security — Xapo

Jan 2018 – May 2020 · Remote

- Led five senior engineers across time zones owning the risk subsystem behind every Xapo transaction.
- Rebuilt fraud detection from scratch across 8 microservices. Average decision time went from about 5 minutes to **under 3 seconds**, with much stronger error tolerance.
- Built an ML-based detection engine that **cut false positives 50%**. Designed a Merkle-tree Proof-of-Reserves architecture on the bitcoin blockchain.

Cofounder & Engineering Lead — Loanzen

Jul 2015 – Jan 2018

Venture-funded fintech lending to India's unbanked SME segment.

- Hired and built the engineering team. Shipped a fully automated React Native loan origination platform with automated disbursal and Aadhaar-signed loan agreements, among the first in India to do so.
- Built the alternative-credit data stack: Android background collectors for SMS and app usage, unsupervised clustering plus a hierarchical naive Bayes classifier over the SMS corpus, declarative PDF parsing for tax documents, bank statement and QuickBooks analysis.

Cofounder & Engineering Lead — VoxApp

Jan 2013 – May 2015 · Remote

Enterprise SaaS survey and mobile-data platform. Hybrid survey app packaged as Android, Chrome desktop and web, with server-side rendering for feature phones and hot-update tooling that avoided full app releases.

Earlier: Cofounder & VP Engineering, GetRightPrice (2012) · Cofounder & Head of Product Development, PriceShred (2011) · Software Developer, Rearden Commerce (2010–11) · Research Engineer Intern, Utopia Compression (2010) · Research, Teamcore Group at USC (2009).

PROJECTS

Harness Engineering (github.com/vertexcover-io/harness-engineering). MIT-licensed Claude Code and Codex plugin: brainstorm, spec, plan, TDD, two-pass review, live end-to-end verification, ship. Persistent artifacts for every run and a DAG dashboard.

AgentLoop / Dispatch (agentloop.vertexcover.io/built). AgentLoop is a daily AI digest built by agents inside that harness, running on Dispatch, the multi-tenant newsletter platform underneath it: 34+ collectors, dedupe, two-stage ranking, human review, then email, web archive, LinkedIn and X from one run.

Strot (github.com/vertexcover-io/strot). API scraper that discovers hidden endpoints and generates the extraction code, with component evals and observability built in.

Tarash Gateway (tarash.vertexcover.io). One typed interface across 11 AI media providers and 100+ models for video, image and audio, with config-driven switching and ordered fallback chains.

SceneFlow (github.com/vertexcover-io/sceneflow). Cut-point selection for AI-generated video: speech boundaries refined against audio energy, then frames ranked on quality and stability so clips do not end mid-word or on a blink.

SKILLS

Agentic delivery: Claude Code, Codex, Cursor, spec-driven development (EARS), prompt and skill versioning, agent evals, quality gates, auditing agent-generated code for hallucinations and security issues.

AI and ML: multi-agent orchestration, RAG, tool calling, model fine-tuning, LLM tracing and observability, guardrails, human-in-the-loop design.

Platform: TypeScript, Python, Go, React, Node/Hono, PostgreSQL, Redis/BullMQ, Docker, AWS, Terraform, CI/CD, GPU and inference infrastructure.

Growth and instrumentation: content pipelines at scale, multi-tenant domains and branding, event analytics, open and click attribution, multi-channel publishing to email, LinkedIn and X.

EDUCATION

MS Computer Science, University of Southern California, Los Angeles

2008 – 2010